

Questions with Answer Keys

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Q1 - Single Correct

Let z be a complex number satisfying $|z^2 + 2z \cos \alpha| \leq 1$. ($\alpha \in R$) then maximum values $|z|$ must be

(1) $\sqrt{2} + 1$

(2) $\sqrt{3} - 1$

(3) $\sqrt{3} + 1$

(4) $\sqrt{6}$

Q2 - Single Correct

If z_1, z_2 and z_3 are three complex numbers and a, b and c are three positive real numbers, such that

$$\frac{a}{|z_2 - z_3|} = \frac{b}{|z_3 - z_1|} = \frac{c}{|z_1 - z_2|}, \text{ then the value of } \frac{a^2}{z_2 - z_0} + \frac{b^2}{z_0 - z} + \frac{c^2}{z} \text{ is}$$

(1) 0

(2) 1

(3) 2

(4) 3

Q3 - Single Correct

Angular hexagon $ABCDEF$ is inscribed in a circle of radius 1. A point P is taken on the circle of radius 2 having same centre. $PA^2 + PB^2 + PC^2 + PD^2 + PE^2 + PF^2$ is equal to

(1) 18

(2) 12

(3) 24

(4) 30

Q4 - Single Correct

If $|z - 1| = 1$ and $\arg z = \theta$ ($z \neq 0$) and $0 < \theta < \frac{\pi}{2}$, then $1 - \frac{2}{z}$ is equal to

(1) $\tan \theta$

(2) $i \tan \theta$

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(3) $\tan \frac{\theta}{2}$

(4) $i \tan \frac{\theta}{2}$

Q5 - Single Correct

If all the three roots of $az^3 + bz^2 + cz + d = 0$ have negative real part ($a, b, c, d \in R$), then

(1) $bc > ad$

(2) $bc < ad$

(3) $bc = ad$

(4) None of these

Q6 - Single Correct

Suppose A be a complex number and $n \in N$, such that $A^n = (A + 1)^n = 1$, then the least value of n is

(1) 3

(2) 6

(3) 9

(4) 12

Q7 - Single Correct

If $1, z_1, z_2, \dots, z_{n-1}$ are the n th roots of unity and ω is a non-real complex cube root of unity, then the product $\prod_{r=1}^{n-1} (\omega - z_r)$ cannot be equal to

(1) 0

(2) 1

(3) -1

(4) $1 + \omega$

Q8 - Single Correct

If $z_r, r = 1, 2, 3, \dots, 50$ are the roots of the equation $\sum_{r=0}^{50} (z)^r = 0$, then the value of $\sum_{r=1}^{50} \frac{1}{z_r - 1}$, is

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- (1) -85
- (2) -25
- (3) 25
- (4) 75

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Q9 - Single Correct

Consider the circle $|z - 5i| = 3$ and two points z_1 and z_2 on it such that $|z_1| < |z_2|$ and $\arg(z_1) = \arg(z_2) = \frac{\pi}{3}$. A tangent is drawn at z_2 to the circle, which cuts the real axis at z_3 , then $|z_3|$ is equal to

- (1) 3
- (2) $\sqrt{3}$
- (3) $\sqrt{7}$
- (4) $\sqrt{11}$

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Q10 - Single Correct

If the complex numbers z_1 and z_2 are the solutions of the equation $z^2 - z + 1 = 0$, then the value of the expression $E = (z_1^4 - z_1^3 + 2z_1^2 - 2z_1 + 1)^{2013} + (z_2^4 - z_2^3 + 2z_2^2 - 2z_2 + 1)^{2013}$, is

- (1) 1
- (2) 2
- (3) -200
- (4) -2

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Q11 - Multiple Correct

Let P be any point on the circle with OP as diameter (O being origin). The points Q and R are on the same side of the diameter such that $\angle POQ = \angle QOR = \theta$. Also P, Q, R be are on the numbers z_1, z_2 and z_3 respectively such that $\sqrt{2}\sqrt{3}z_2^2 = (1 + \sqrt{3})z_1z_3$, then θ

- (1) $\frac{\pi}{8}$

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(2) $\frac{\pi}{12}$

(3) $\frac{\pi}{18}$

(4) $\frac{\pi}{10}$

Q12 - Multiple Correct

One vertex of the triangle of maximum area that can be inscribed in the curve $|z - 2i| = 2$, is $2 + 2i$ remaining vertices is/are

(1) $-1 + i(2 + \sqrt{3})$

(2) $-1 - i(2 + \sqrt{3})$

(3) $-1 + i(2 - \sqrt{3})$

(4) $-1 - i(2 - \sqrt{3})$

Q13 - Multiple Correct

If z is a complex number which simultaneously satisfy the equations $3|z - 12| = 5|z - 8i|$ and

$|z - 4| = |z - 8|$, then the $\text{Im}(z)$ can be

(1) 15

(2) 16

(3) 17

(4) 8

Q14 - Multiple Correct

If z satisfy the inequality $|z - 1 - 2i| \leq 1$, then

(1) $\min\{\arg(z)\} = \tan^{-1}\left(\frac{3}{4}\right)$

(2) $\max\{\arg(z)\} = \frac{\pi}{2}$

(3) $\min(|z|) = \sqrt{5} - 1$

(4) $\max(|z|) = \sqrt{5} + 1$

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Q15 - Multiple Correct

If z is a complex number then the equation $z^2 + z|z| + |z|^2 = 0$ is satisfied by

- (1) $z = kw, k \in R$
- (2) $z = kw^2$ where k is non-negative real
- (3) $z = kw$, where k is positive real
- (4) $z = k\omega^2, k \in R$

Q16 - Multiple Correct

Let the complex number $z = x + iy$ (where $x, y \in R$) satisfy $\cot^{-1}(\log_3 |2z + 1|) > \cot^{-1}(\log_3 |2z - 1|)$, then x can be

- (1) -2
- (2) -1
- (3) 1
- (4) 2

Q17 - Multiple Correct

If $|z - 2| = \min\{|z|, |z - 4|\}$, then possible value (s) of $\text{Re}(z)$ can be

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q18 - Multiple Correct

Let z_1, z_2 and z_3 be the coordinates of the vertices of the $\Delta A_1 A_2 A_3$, then which of the following statements are equivalent?

- (1) $A_1 A_2 A_3$ is an equilateral triangle
- (2) $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$, where ω is a cube root of unity

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(3) $\frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_2}{z_1 - z_3}$

(4) $\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \end{vmatrix} = 0$

Q19 - Multiple Correct

If $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the imaginary n th roots of unity, then the product $\prod_{r=1}^{n-1} (i - \alpha_r)$ (where, $i = \sqrt{-1}$) can take the value(s)

(1) 0

(2) 1

(3) i

(4) $(1 + i)$

Q20 - Multiple Correct

If $z \in \mathbb{C}$, then which of the following relation(s) represents a circle on an argand diagram?

(1) $|z - 1| + |z + 1| = 3$

(2) $(z - 3 + i)(\bar{z} - 3 - i) = 5$

(3) $3|z - 2 + i| = 7$

(4) $|z - 3| = 2$

Q21 - Multiple Correct

Let z_1, z_2 and z_3 be three complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$.

Then, $|z_1 + z_2 + z_3|$ can take the value equals

(1) 1

(2) 2

(3) 3

(4) 4

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Q22 - Multiple Correct

Let A and B be two distinct points denoting the complex numbers α and β , respectively. A complex number z lies between A and B such that $z \neq \alpha, z \neq \beta$. Which of the following relation(s) hold good?

(1) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

(2) There exists a positive real number t such that $z = (1 - t)\alpha + t\beta$

(3) $\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$

(4) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

Q23 - Multiple Correct

The equation $|z - i| + |z + i| = k, k > 0$ can represent

(1) an ellipse, if $k > 2$

(2) line segment if, $k = 2$

(3) an ellipse, if $k = 5$

(4) line segment, if $k = 1$

Q24 - Multiple Correct

The equation $||z + i| - |z - i|| = k$ represents

(1) a hyperbola, if $0 < k < 2$

(2) a pair of ray, if $k > 2$

(3) a straight line, if $k = 0$

(4) a pair of ray, if $k = 2$

Q25 - Multiple Correct

If POQ is a straight line through the origin O . P and Q represent the complex numbers $a + ib$ and $c + id$ respectively and $OP = OQ$. Then,

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(1) $|a + ib| = |c + id|$

(2) $a + c = b + d$

(3) $a + b = c + d$

(4) None of these

Q26 - Comprehension Type

Passage I (For Question 26, 27)

Let z_1 and z_2 be two complex numbers satisfying the equation $3z_1^2 - 2z_1z_2 + 2z_2^2 = 0$ and also $\operatorname{Re}\left(\frac{z-2}{z+2}\right) = 0$

. Also, P_1, P_2 and O are points in complex plane corresponding to z_1, z_2 and origin respectively. Then,

The value of $|z_1|$ is equal to

(1) 1

(2) $\sqrt{5}$

(3) 2

(4) $2\sqrt{2}$

Q27 - Comprehension Type

Area of $\triangle OP_1P_2$ is

(1) $\sqrt{5}$

(2) $2\sqrt{5}$

(3) $3\sqrt{5}$

(4) $4\sqrt{5}$

Q28 - Comprehension Type

Passage II (For Question 28, 29)

Let $A(z_1), B(z_2)$ and $C(z_3)$ be the vertices of $\triangle ABC$ such that $z_3 + i\omega z_2 = (1 + i\omega)z_1$, where $\omega (\neq 1)$ be a cube root of unity, then

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 $\triangle ABC$ is

(1) right angled isosceles

(2) isosceles

(3) scalene

(4) equilateral

Q29 - Comprehension Type

Coordinates of circumcentre of $\triangle ABC$ is

(1) $\frac{(z_1 - z_2) + (\omega - 4)z_3}{\omega}$

(2) $\frac{\omega(z_1 - z_2) + (\omega + 4)z_3}{\omega}$

(3) $\frac{\omega(z_1 - z_2) + z_3}{\omega}$

(4) $\frac{(1 + \omega)z_2 - z_3}{\omega}$

Q30 - Comprehension Type

Passage III (For Question 30, 31)

Consider $z = x + iy$, where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$. Let $A(z_0)$ be a complex number satisfying $|z - 2i| = \operatorname{Im}(z)$ and $\arg(z_0 + 1) = \alpha$, thenIf α is minimum, then

(1) $\frac{2i - z_0}{1 + 2i} + \frac{\bar{z}_0 + 2i}{2i - 1} = 0$

(2) $\operatorname{Im}\left(\frac{2i - z_0}{1 + 2i}\right) = 0$

(3) $\operatorname{Re}\left(\frac{2i + z_0}{1 + 2i}\right) = 0$

(4) $\frac{2i - z_0}{1 + 2i} - \frac{\bar{z}_0 + 2i}{2i - 1} = 0$

Q31 - Comprehension Type

If α is minimum, then the circle $\left|z - \left(\frac{z_0 + 2i}{2}\right)\right| = \frac{|z_0 - 2i|}{2}$ contains points whose

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(1) $\operatorname{Im}(z) \leq 0$

(2) $\operatorname{Re}(z) \geq 1$

(3) $\operatorname{Re}(z) \leq 0$

(4) $\operatorname{Im}(z) \geq 1$

Q32 - Comprehension Type

Passage IV (For Question 32, 33)

Let $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ be the vertices of a trapezium in an argand plane. Let

$|z_1 - z_2| = 4$, $|z_3 - z_4| = 10$ and diagonals AC and BD intersect at P . it is given that

$$\arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2} \text{ and } \arg\left(\frac{z_3 - z_2}{z_4 - z_1}\right) = \frac{\pi}{4}$$

Area of the trapezium $ABCD$ is equal to

(1) $\frac{130}{3}$

(2) $\frac{160}{3}$

(3) $\frac{190}{3}$

(4) $\frac{140}{3}$

Q33 - Comprehension Type

Area of $\triangle PCB$ is equal to

(1) $\frac{100}{21}$

(2) $\frac{200}{21}$

(3) $\frac{100}{7}$

(4) $\frac{400}{21}$

Q34 - Comprehension Type

Passage V (For Question 34, 35)

Let A , B and C be three sets of complex numbers as defined below

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$$A = \{z : |z + 1| \leq 2 + \operatorname{Re}(z)\}, \quad B = \{z : |z - 1| \geq 1\} \text{ and } C = \left\{z : \left|\frac{z+1}{z-1}\right| \geq 1\right\}$$

The number of points having integral coordinates in the region $A \cap B \cap C$ is

- (1) 4
- (2) 5
- (3) 6
- (4) 10

Q35 - Comprehension Type

The area of the region $A \cap B \cap C$ is

- (1) $2\sqrt{3}$
- (2) $\sqrt{3}$
- (3) $4\sqrt{3}$
- (4) 2

Q36 - Matrix Match (Only one option is correct)

If $x = z_1 + z_2 + z_3, y = z_1 + z_2\omega + z_3\omega^2, z = z_1 + z_2\omega^2 + z_3\omega$, where ω is the cube root of unity, then

	Column I		Column II
i.	$ x ^2$	p.	$\Sigma z_1 ^2 + \bar{z}_1(z_2\omega + z_3\omega^2) + \bar{z}_2(z_3\omega + z_1\omega^2) + \bar{z}_3(z_1\omega + z_2\omega^2)$
ii.	$ y ^2$	q.	$\Sigma z_1 ^2 + \bar{z}_1(z_2\omega^2 + z_3\omega) + \bar{z}_2(z_3\omega^2 + z_1\omega) + \bar{z}_3(z_1\omega^2 + z_2\omega)$
iii.	$ z ^2$	r.	$3\Sigma z_1 ^2$
iv.	$\Sigma x ^2$	s.	$\Sigma z_1 ^2 + \bar{z}_1(z_2 + z_3) + \bar{z}_2(z_3 + z_1) + \bar{z}_3(z_1 + z_2)$
		t.	$(z_1 + z_2 + z_3)^2$

(1)	(i)	(ii)	(iii)	(iv)
	s	p	q	r
(2)	(i)	(ii)	(iii)	(iv)
	s	q	p	r

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(3)

(i)	(ii)	(iii)	(iv)
r	p	q	s

(4)

(i)	(ii)	(iii)	(iv)
p	q	r	p

Q37 - Matrix Match (Only one option is correct)

Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$.

	Column I		Column II
i.	Maximum value of $ z_1 + z_2 $	p.	3
ii.	Minimum value of $ z_1 - z_2 $	q.	1
iii.	Minimum value of $ 2z_1 - 3z_2 $	r.	4
iv.	Maximum value of $ z_1 + 2z_2 $	s.	5

(1)

(i)	(ii)	(iii)	(iv)
q	r	p	s

(2)

(i)	(ii)	(iii)	(iv)
p	q	r	s

(3)

(i)	(ii)	(iii)	(iv)
r	p	s	q

(4)

(i)	(ii)	(iii)	(iv)
s	q	p	r

Q38 - Matrix Match (Only one option is correct)

	Column I		Column II
i.	$z^2 - z + 1 = 0$	p.	$\arg(z) = -\frac{2\pi}{3}$
ii.	$z^2 + z + 1 = 0$	q.	$\arg(z) = -\frac{\pi}{3}$
iii.	$2z^2 + 1 + i\sqrt{3} = 0$	r.	$\arg(z) = \frac{\pi}{3}$
iv.	$2z^2 + 1 - i\sqrt{3} = 0$	s.	$\arg(z) = \frac{2\pi}{3}$

(1)

(i)	(ii)	(iii)	(iv)
q r	p s	q s	p r

(2)

(i)	(ii)	(iii)	(iv)
q r s	p s	q	p r

Questions with Answer Keys

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(3)	(i)	(ii)	(iii)	(iv)
	p q	p s	q s	p s

(4)	(i)	(ii)	(iii)	(iv)
	q r	p q	q s	p r

Q39 - Integer Type

A positive integer n is randomly guessed. Let the probability that $x^{n+1} - x^n + 1$ is divisible by $x^2 - x + 1$ be P . The value of $12P$ is

Q40 - Integer Type

Let z and w be complex numbers such that $z + w = i$ and $z^2 + w^2 = 1$. If area of triangle formed by z , w and origin is equal to A , if the value of $80A^2$ is 1.5×10^x . Find the value of x .

Q41 - Integer Type

If the roots of the equation $z^4 + az^3 + (12 + 9i)z^2 + bz = 0$ are the vertices of a square, if the value of $2b - 13a$ is $x \times 10$, find the value of x .

Q42 - Integer Type

A region S in complex plane is $S = \{x + iy, -1 \leq x \leq 1 \text{ and } -1 \leq y \leq 1\}$. A complex number $z = x + iy$ is chosen uniformly at random from S . If the probability P that the complex number $\frac{3}{4}(1 + i)z$ is also in S can be expressed in lowest form $\frac{m}{n}$, then find number of digits in $(m + n)$.

Q43 - Integer Type

Let M be the set of all complex number z that satisfy $|z - 4 - 3i| = r$. If r is a positive constant such that there is only one real number, which belongs to M . For the value of r , find the maximum value of $|z|$.

Q44 - Integer Type

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Let z be a complex number satisfying $|2z + 3i| = |z^2|$. If M and m are the maximum and minimum value of z , then find the value of $(M + m)$

Q45 - Integer Type

Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$. Find the number of elements in the range of $|z_1 + z_2 + z_3|$.

Q46 - Integer Type

Let w be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$, then the number of distinct complex number z satisfying

$$\begin{vmatrix} z+1 & w & w^2 \\ w & z+w^2 & 1 \\ w^2 & 1 & z+w \end{vmatrix} = 0, \text{ is equal to}$$

Q47 - Integer Type

Let $w = e^{2\pi/3}$ and a, b, c, x, y, z be non-zero complex numbers such that $a + b = c = x$, $a + bw + cw^2 = y$ and $a + bw^2 + cw = z$. Then, the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is

Q48 - Integer Type

The minimum value of $|1 + z| + |1 - z|$, where z is a complex number, is

Q49 - Integer Type

The system of equation $\left. \begin{matrix} |z + 1 - i| = 2 \\ \operatorname{Re}(z) \geq 1 \end{matrix} \right\}$, where z is a complex number, has number of solutions

Q50 - Integer Type

If z is a complex number, then the minimum value of $|z - 1|^2 + |z - 2|^2 + |z - 3|^2$, is equal to

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Answer Key					
// mathongo	// mathongo	// mathongo	// mathongo	// mathongo	// mathongo
Q1 (1)	Q2 (1)	Q3 (4)	Q4 (2)		
// mathongo	// mathongo	// mathongo	// mathongo	// mathongo	// mathongo
Q5 (1)	Q6 (2)	Q7 (3)	Q8 (2)		
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Q9 (4)	Q10 (2)	Q11 (2)	Q12 (1,3)		
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Q13 (3,4)	Q14 (1,2,3,4)	Q15 (2,3)	Q16 (1,2)		
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Q17 (1,3)	Q18 (1,2,3,4)	Q19 (1,2,3,4)	Q20 (2,3,4)		
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Q21 (1,2)	Q22 (1,2,3,4)	Q23 (1,2,3)	Q24 (1,3,4)		
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Q25 (1,2)	Q26 (3)	Q27 (1)	Q28 (2)		
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Q29 (4)	Q30 (1)	Q31 (4)	Q32 (4)		
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Q33 (2)	Q34 (2)	Q35 (1)	Q36 (1)		
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Q37 (2)	Q38 (1)	Q39 (2)	Q40 (1)		
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Q41 (6)	Q42 (2)	Q43 (8)	Q44 (4)		
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Q45 (2)	Q46 (1)	Q47 (3)	Q48 (2)		
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Q49 (1)	Q50 (2)				
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